Keywords: multimodal diffusion LMs • hallucination • visual grounding • training-free inference

VIGOR: Visual Gain Ordering for Hallucination Mitigation in Multimodal Discrete Diffusion Language Models

Reordering Unmasking by Counterfactual Visual Evidence Consistently Reduces Hallucination Without Any Fine-Tuning

NeurIPS 2026 Submission
arXiv:2611.05066 • cs.CV / cs.CL • NeurIPS 2026

Multimodal discrete diffusion LMs hallucinate because language priors dominate unmasking confidence, causing tokens to be committed before visual evidence can influence them. VIGOR reorders unmasking by visual gain—the difference in confidence between normal and visually-ablated inference passes—ensuring image-grounded tokens are committed first. Applied to LLaDA-V and Lumina-DiMOO, it consistently reduces hallucination with no training required.

AT A GLANCE

Problem: Hallucination in multimodal dLLMs stems partly from premature commitment: positions are unmasked by language-prior confidence before visual evidence is fully integrated.

Why it matters: Multimodal dLLMs are emerging alternatives to autoregressive VLMs; their hallucination problem must be solved for practical deployment.

Method: At each denoising step, compare confidence under normal and visually-ablated passes; unmask in order of visual gain (the difference).

Result: Consistent hallucination reduction and improved multimodal reasoning on both LLaDA-V and Lumina-DiMOO, across all evaluated benchmarks.

Evidence: POPE, MMHal-Bench, NocCaps, and standard VQA benchmarks.

The Big Picture

masked-token prediction: starting from a fully masked sequence, the model at each step selects the positions it is most confident about and reveals them. When paired with a vision encoder, these models become multimodal discrete diffusion LMs (multimodal dLLMs) capable of generating text descriptions conditioned on images.

In a masked dLLM, the unmasking order is flexible: positions can be committed in any sequence. The model currently commits positions in order of raw confidence—but high-confidence positions are often those that fit well in the sentence based on language priors alone, independent of the image. VIGOR identifies this as the *premature commitment problem*: by committing language-prior-driven positions first, the model forecloses visual evidence before it can influence downstream positions. VIGOR's solution is to change the commitment order so that visually-grounded positions are committed first, providing a visual anchor for subsequent language-driven completions.

Why Existing Approaches Fall Short

Raw confidence ordering (the standard strategy in masked dLLMs) conflates language-prior confidence and visual confidence, unmasking high-language-prior positions first regardless of their visual grounding.

Contrastive decoding methods for autoregressive VLMs (e.g., VCD, ICD) subtract language-only scores from image-conditioned scores at the token level, but were designed for per-token generation in autoregressive models and do not transfer cleanly to the unmasking-order decision in dLLMs.

Training-based approaches (RL fine-tuning, reinforcement from visual feedback) can reduce hallucination but require additional compute and do not apply to deployed frozen models. VIGOR is the first method to apply contrastive visual reasoning to the *commitment order decision* in masked dLLMs, enabling training-free hallucination mitigation specific to the dLLM generation structure.

Core Mechanism

At denoising step t, let m_t be the current masking state, I the image, and T the text context. For each masked position $j \in m_t$, VIGOR computes:

1. **Normal confidence:** $P(j|I,T) = \max_v p_\theta(v|I,T,m_t,j)$
2. **Ablated confidence:** $P(j|\varnothing,T) = \max_v p_\theta(v|\varnothing,T,m_t,j)$
3. **Visual gain:** $VG(j) = P(j|I,T) - P(j|\varnothing,T)$

VIGOR selects the top-k positions by visual gain to unmask at step t, rather than the top-k by raw confidence. The null image embedding is a fixed zero vector in visual token space.

at position j under the image-conditioned model. The visual gain is:

$$VG(j) = p_\theta(\hat{y}_j | I, T, m_t, j) - p_\theta(\hat{y}_j | \varnothing, T, m_t, j)$$

Positions with $VG(j) > 0$ have their most likely token more supported by the image than by language priors alone. The unmasking set at step t is $U_t = \text{arg top-}k_{j \in m_t}\ VG(j)$. The token values assigned to revealed positions are taken from the image-conditioned pass.

Learning or Inference Procedure

VIGOR is a pure inference-time method. Required inputs: a pretrained multimodal dLLM (LLaDA-V, Lumina-DiMOO, or any masked-diffusion VLM with accessible token logits), a null image embedding (zero vector in visual token space), and two forward passes per denoising step (normal and ablated). The added cost is one extra forward pass per step. For a standard denoising procedure with $T_d = 10$ steps, VIGOR adds 10 forward passes over baseline, roughly doubling inference cost.

What the Guarantee Says

No formal guarantee is claimed. The empirical claim is: on all evaluated hallucination benchmarks, VIGOR reduces hallucination rates compared to raw-confidence unmasking ordering. Visual gain is a better proxy for visual grounding than raw confidence, and this better proxy translates to fewer committed hallucinations.

Experimental Findings

8–14% (absolute) on both LLaDA-V and Lumina-DiMOO; largest gains on adversarial POPE splits.

- **MMHal-Bench:** Hallucination score improves by 0.4–0.7 points (on a 0–4 scale) on both backbones.
- **NocCaps:** CIDEr score improves by 3–5 points; caption specificity increases measurably.
- **Standard VQA:** Performance is unchanged (±0.5%), confirming the reordering does not hurt non-hallucination-sensitive tasks.
- **Inference cost:** Roughly 2× baseline; latency is acceptable for batch inference applications.

Ablations and Interpretation

Null image type: Zero vector vs. learned null token vs. Gaussian noise all produce similar visual gain scores; zero vector is used for simplicity.

Top-k selection vs. threshold: Using $VG(j) > 0$ as a threshold occasionally unmasks too few or too many positions per step; fixed-k matching the base model's schedule is preferred.

Position of VIGOR in the denoising schedule: Applying VIGOR only in the first half of denoising steps captures 80% of the hallucination reduction at half the extra cost, confirming that early commitment decisions are the most consequential.

Comparison with token-level contrastive decoding: Applying VCD-style contrastive decoding to the token values (not the commitment order) gives smaller gains than VIGOR, confirming that the commitment-order effect is the primary mechanism.

Reference

NeurIPS 2026 Submission. **VIGOR: Visual Gain Ordering for Hallucination Mitigation in Multimodal Discrete Diffusion Language Models.** arXiv:2611.05066. NeurIPS 2026. arxiv.org/abs/2611.05066